

E.G.S. PILLAY ENGINEERING COLLEGE

(Autonomous)

NAGAPATTINAM – 611 002.

(Affiliated to Anna University, Chennai | Accredited by NAAC with 'A++' Grade
Accredited by NBA | Approved by AICTE, New Delhi)



B.E ELECTRICAL AND ELECTRONICS ENGINEERING (R-2023)

CURRICULUM AND SYLLABUS FOR SECOND YEAR: THIRD SEMESTER (III)

Course Code	Course Name	L	T	P	C	Maximum Marks		
						CA	ES	Total
Theory Course								
2301MA305	Engineering Mathematics-III (Transforms & Optimization)	3	2	0	4	40	60	100
2302EE301	Electrical Machines-I	3	2	0	4	40	60	100
2302EE302	Analog Electronics	3	0	0	3	40	60	100
2302EE303	Digital Electronics	3	0	0	3	40	60	100
Theory Cum Laboratory Course								
2302EE304	Measurements & Instrumentation	3	0	2	4	50	50	100
2301GEX07	Environmental Science and sustainability	1	0	2	2	50	50	100
Laboratory Course								
2302EE351	Electrical Machines–I Laboratory	0	0	2	1	60	40	100
2302EE352	Analog and Digital Electronics Laboratory	0	0	2	1	60	40	100
2304GE301	Professional Development Course – I	0	0	2	1	100	–	100
2301LS301	Life Skills - III	0	0	0	0	100	–	100
Total		16	04	10	23	580	420	1000

L – Lecture | T – Tutorial | P – Practical | CA – Continuous Assessment | ES – End Semester

MODULE III	Z –TRANSFORMS	9 Hours
Z- Transforms - Elementary properties – Inverse Z - transform (using partial fraction and residues) – Convolution theorem - Formation of difference equations – Solution of difference equations using Z – transform.		
MODULE IV	INTRODUCTION TO WAVELETS TRANSFORMS	9 Hours
The origins of wavelets, Wavelets and other wavelet like transforms, Continuous Wavelet Transform: Continuous time-frequency representation of signals, Properties of wavelets used in continuous wavelet transform, Continuous versus discrete wavelet transform. Discrete Wavelet Transform: Haar scaling Functions and function spaces, Translation and scaling of $\phi(t)$.		
MODULE V	INTRODUCTION TO OPTIMIZATION	9 Hours
Introduction to Optimization: Engineering applications of Optimization – Statement of an Optimization problem – Optimal Problem formulation – Classification of Optimization problem. Optimum design Concepts: Definition of Global and Local optima – Optimality criteria – Review of basic calculus concepts – Global optimality.		
TOTAL: 45 + 15 = 60 HOURS		
REFERENCES:		
1. Grewal B.S., 41st Edition, 2011, “Higher Engineering Mathematics”, Khanna Publishers, New Delhi.		
2. Ramana B.V., 11th Reprint, 2010, “Higher Engineering Mathematics”, Tata McGraw Hill Co. Ltd., NewDelhi		
3. Veerarajan T., Engineering Mathematics for first year, Tata McGraw-Hill, New Delhi, 2018.		
4. K. P. Soman, K. I. Rmachandran, N. G. Resmi, “Insight into Wavelets: From Theory to Practice, (Third Edition)”, PHI Learning Pvt. Ltd., 2010.		
5. C. S. Burrus, Ramose and A. Gopinath, Introduction to Wavelets and Wavelet Transform, Prentice Hall Inc.		
6. Rao S. S. - „Engineering Optimization, Theory and Practice” - New Age International Publishers - 2012 - 4th Edition.		
7. Arora J. - „Introduction to Optimization Design” - Elsevier Academic Press, New Delhi - 2004		
8. http://www.nptelvideos.in/2012/12/probability-random-variables.html (Link for NPTEL/SWAYAM/MOOC Courses)		
9. https://matlabacademy.mathworks.com/details/introduction-to-symbolic-math-with-matlab/symbolic (Link for modern tool usage)		
10. Grewal B.S., 41st Edition, 2011, “Higher Engineering Mathematics”, Khanna Publishers, New Delhi.		
11. Ramana B.V., 11th Reprint, 2010, “Higher Engineering Mathematics”, Tata McGraw Hill Co. Ltd., NewDelhi		

2302EE301	ELECTRICAL MACHINES - I	L	T	P	C																																																																								
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PREREQUISITE:																																																																													
1. Electric circuit analysis																																																																													
COURSE OBJECTIVES:																																																																													
1. To understand the concept of electromechanical energy conversion system in DC machines.																																																																													
2. To identify the appropriate test to determine the performance parameters of a given machine.																																																																													
3. To deliberate the working of single and three phase transformers.																																																																													
COURSE OUTCOMES:																																																																													
On the successful completion of the course, students will be able to																																																																													
CO1:	Evaluate the behavior of DC generators under various phenomena such as armature reaction and commutation.																																																																												
CO2:	Assess the operation and characteristics of DC motor with appropriate control strategies.																																																																												
CO3:	Compute the efficiency and regulation of DC machines by conducting various test.																																																																												
CO4:	Develop the equivalent circuit of single-phase transformer.																																																																												
CO5:	Elucidate the operation, applications of three phase and special purpose transformers.																																																																												
COs Vs. POs MAPPING:																																																																													
<table><tr><th>COs</th><th>PO1</th><th>PO2</th><th>PO3</th><th>PO4</th><th>PO5</th><th>PO6</th><th>PO7</th><th>PO8</th><th>PO9</th><th>PO10</th><th>PO11</th></tr><tr><td>CO1</td><td>3</td><td>3</td><td>2</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO2</td><td>3</td><td>3</td><td>2</td><td>1</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO3</td><td>3</td><td>3</td><td>2</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO4</td><td>3</td><td>3</td><td>2</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr><tr><td>CO5</td><td>3</td><td>2</td><td>1</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td><td>-</td></tr></table>						COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	CO1	3	3	2	1	1	-	-	-	-	-	-	CO2	3	3	2	1	1	-	-	-	-	-	-	CO3	3	3	2	1	-	-	-	-	-	-	-	CO4	3	3	2	1	-	-	-	-	-	-	-	CO5	3	2	1	-	-	-	-	-	-	-	-
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11																																																																		
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CO4	3	3	2	1	-	-	-	-	-	-	-																																																																		
CO5	3	2	1	-	-	-	-	-	-	-	-																																																																		
COs Vs. PSOs MAPPING:																																																																													
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CO5	3	-																																																																											
COURSE CONTENTS:																																																																													
MODULE I	DC GENERATOR				12 Hours																																																																								
Fundamentals of Magnetic circuits - Statically and dynamically induced EMF; Principle of operation-constructural details- Single and double layer winding -Types -EMF equation-armature reaction-commutation-methods of improving commutation; OCC and load characteristics; Applications; Simulation Studies on DC generator.																																																																													
MODULE II	DC MOTOR				12 Hours																																																																								
Principle of operation - Significance of back EMF; Torque equations and power developed by armature; Characteristics of DC motor; starters; Speed control methods; Applications of DC motors; Simulation studies on DC motor.																																																																													

MODULE III	TESTING OF DC MACHINES	12 Hours
Losses and efficiency in DC machine - Condition for maximum efficiency; Testing of DC Machines-Brake test, Swinburne's test, Hopkinson's test, Field test, Retardation test, Separation of core losses.		
MODULE IV	SINGLE PHASE TRANSFORMER	12 Hours
Construction and principle of operation- EMF equation- Equivalent circuit- ; Phasor diagrams- Testing of transformer - Polarity test, Open and short circuit tests, Sumpners test, Efficiency and regulation, All day efficiency; Parallel operation; Applications of single-phase transformer.		
MODULE V	THREE PHASE & SPECIAL PURPOSE TRANSFORMERS	12 Hours
Three phase transformer – Construction-Connection; Zig-Zag Transformer; Autotransformer - Copper saving; Power transformer -Types; Distribution transformer- Construction- Roles; Instrument transformer - Role of CT & PT in power system protection; Welding Transformers, Traction Transformers, Pulse Transformers, Energy efficient transformer.		
TOTAL: 60 HOURS		
REFERENCES:		
1. D. P. Kothari and I. J. Nagrath, <i>Electric Machines</i> , Tata McGraw Hill Publishing Company Ltd, 2010.		
2. J.B. Gupta, <i>Theory & Performance of Electrical Machines</i> , Kataria, S. K., & Sons, 2013.		
3. Edward Hughes, <i>Electrical and Electronic Technology</i> , 12 th edition, Pearson, 2016.		
4. P. S. Bimbhra, <i>Electrical Machinery</i> , Khanna Publishers, 7th edition, 2011.		
5. B. L. Theraja and A. K. Theraja, — <i>Text Book of Electrical Technology: AC & DC Machines (Volume- 2)</i> , S.Chand& Company Ltd., New Delhi, 2008.		
6. M.N.Bandyopadhyay, <i>Electrical Machines Theory and practice</i> , PHI Learning Pvt. Ltd, New Delhi 2007.		
7. http://nptel.ac.in/courses/117106086/		

MODULE II	BIPOLAR JUNCTION TRANSISTOR AND CIRCUITS	9 Hours
BJT: Structure- Operation- Biasing circuits; V-I characteristics in common base, common emitter and common collector configurations; DC and AC load line analysis- Determination of Q point; Small signal model; Analysis and comparison of CB, CE and CC amplifiers.		
MODULE III	FIELD EFFECT TRANSISTOR AND CIRCUITS	9 Hours
JFET: Structure- Operation - n channel and p channel; V-I characteristics and biasing circuits of JFET.		
MOSFET: Structure and operation of D-MOSFET & E-MOSFET; V-I characteristics- Biasing circuits- small signal model; Analysis of common source and common drain amplifiers.		
MODULE IV	DIFFERENTIAL AND POWER AMPLIFIERS	9 Hours
Differential Amplifier: Common mode and difference mode analysis of BJT based differential amplifier.		
Power Amplifiers: Class A, class B, class C and class AB Amplifiers.		
MODULE V	OSCILLATORS	9 Hours
Basics about feedback, Barkhausen criterion; RC oscillators – RC phase shift and Wien bridge oscillators; LC oscillators – Hartley, Colpitts and Clapp; Crystal oscillators – Miller and Pierce crystal oscillators.		
TOTAL: 45 HOURS		
REFERENCES:		
1. Milman, Halkias and Satyabrata Jit, <i>Electronic Devices and Circuits</i> , McGraw Hill Education (India) Private Ltd., 4th Edition, 2015.		
2. Robert L. Boylestad and Louis Nashelsky, <i>Electronic Devices and Circuit Theory</i> , PHI Ltd., 11th Edition, 2015.		
3. David A. Bell, <i>Electronic Devices and Circuits</i> , Oxford University Press, 5th Edition, 2008.		
4. Thomas L. Floyd, <i>Electronic Devices</i> , An Imprint of Mc Millan publishing company, 10th Edition, 2017.		
5. Prof.A.N.Chandorkar, IIT Bombay online lecture series on Analog Electronics http://nptel.ac.in/courses/117101106/		
6. Prof. S.Karmalkar, IIT Madras, online lecture series on Solid State Devices. http://nptel.ac.in/courses/117106091/		
7. Albert Malvino and David Bates, <i>Electronic Principles</i> , 8th Edition,		
8. https://onlinecourses.nptel.ac.in/noc24_ee69/preview		
9. https://onlinecourses.swayam2.ac.in/nou24_ec04/preview		
10. http://vlabs.iitkgp.ac.in/psac/newlabs2020/vlabiitkgpAE/exp6/index.html		

2302EE303	DIGITAL ELECTRONICS				L	T	P	C																																																																											
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PREREQUISITE:																																																																																			
	1. Applied Physics for Engineers																																																																																		
COURSE OBJECTIVES:																																																																																			
	1. To study the fundamentals of digital systems, programmable logic devices and logic families.																																																																																		
	2. To design and analyze digital systems.																																																																																		
	3. To apply the digital simulation techniques for application oriented digital circuits.																																																																																		
COURSE OUTCOMES:																																																																																			
On the successful completion of the course, students will be able to																																																																																			
CO1	Solve digital system problems using number systems, binary codes, logic gates, Boolean algebra and Karnaugh Map																																																																																		
CO2	Construct combinational logic circuits using logic gates and multiplexers																																																																																		
CO3	Build synchronous sequential logic circuits using excitation table and state diagrams																																																																																		
CO4	Construct asynchronous sequential logic circuits using flow table, transition table, state assignment and state reduction techniques																																																																																		
CO5	Implement Boolean functions and combinational logic circuits using memories, programmable logic devices and logic families																																																																																		
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COURSE CONTENTS:																																																																																			
MODULE I	NUMBER SYSTEMS AND DIGITAL LOGIC FAMILIES								9 Hours																																																																										
Number system; Error detection, corrections & codes conversions; Boolean algebra- De-Morgan’s theorem; switching functions and Simplification using K-maps-3, 4 and 5 variables. Digital Logic Families - comparison of RTL, DTL, TTL, ECL and MOS families; Operation and characteristics of digital logic family.																																																																																			

MODULE II	COMBINATIONAL LOGIC CIRCUIT	9 Hours
Design of adders, subtractors, comparators, code converters, encoders, decoders, multiplexers and demultiplexers. Function realization using multiplexers; Booth multiplier and Array Multiplier; Simulation of simple logic circuits.		
MODULE III	SYNCHRONOUS SEQUENTIAL LOGIC CIRCUITS	9 Hours
Latches- Operation of SR and gated SR latch; Flip flops – Method of edge triggering, SR, JK, Master Slave JK, D and T flip flops; Important signals of FF. Design of Synchronous sequential circuits- Model Selection- State transition diagram- State synthesis table - Design equations-State reduction technique and Implementation; Binary counters-4 bit UP, DOWN and UP/DOWN counters; BCD counters, Ring counters, Johnson counters, shift registers.		
MODULE IV	ASYNCHRONOUS SEQUENTIAL LOGIC CIRCUITS	9 Hours
Design of asynchronous sequential circuits-Design steps- State transition diagram- State table- FF transition table- K-map based Primitive table- State reduction techniques- State assignment and design equations; Races and hazards.		
MODULE V	MEMORY DEVICES, PROGRAMMABLE LOGIC DEVICES AND VHDL	9 Hours
Programmable Logic Devices – PLA, PAL, PLD; Memories: ROM, PROM, EPROM; Introduction to V HDL-Digital design process flow using HDL- Basic VHDL Programming.		
TOTAL: 45 HOURS		
REFERENCES:		
1. M. Morris Mano, — <i>Digital Logic and Computer Design</i> ll, Prentice Hall of India, 4th edition, 2013.		
2. A.Anandkumar, — <i>Fundamentals of digital circuits</i> ll, 3 rd Edition, PHI Learnings Pvt. Ltd, 2014.		
3. Malvino and Leach, <i>Digital Principles and Applications</i> , Tata McGraw Hill, New Delhi, 7th edition, 2011.		
4. Floyd, <i>Digital Fundamentals</i> , Pearson Education, 10th edition, 2011.		
5. John F.Wakerly, <i>Digital Design Principles and Practice</i> , Pearson Education, 4th edition, 2008.		
6. http://nptel.ac.in/courses/117106086/		

2302EE304	MEASUREMENTS AND INSTRUMENTATION				L	T	P	C				
					3	0	2	4				
PREREQUISITE:												
	1. Electric Circuit Analysis											
	2. Engineering Exploration											
COURSE OBJECTIVES:												
	1. To learnt the measuring instrument characteristics and also to calculate different parameters of Instruments.											
	2. To empower students to understand the working of electrical equipment used in everyday life.											
	3. To describe the working principle, selection criteria and applications of various transducers used in measurement systems.											
COURSE OUTCOMES:												
On the successful completion of the course, students will be able to												
CO1	Describe the basic functional elements of measuring instruments and the errors in the measurements systems.											
CO2	Discuss the operation and applications of measuring instrument under typical environment.											
CO3	Identify the unknown values of resistor, inductor and capacitor of given network using suitable bridge circuit.											
CO4	Explain the construction and working principle of various storage and display devices.											
CO5	Make use of sensor and transducers in measuring purpose using data acquisition system.											
COs Vs. POs MAPPING:												
	COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
	CO1	3	3	3	-	-	-	-	-	-	-	1
	CO2	2	2	2	-	-	-	-	-	-	-	1
	CO3	3	3	2	-	3	-	-	-	-	-	1
	CO4	2	3	2	-	-	-	-	-	-	-	1
	CO5	3	3	3	-	-	-	-	-	-	-	1
COs Vs. PSOs MAPPING:												
	COs	PSO1	PSO2									
	CO1	3	3									
	CO2	3	2									
	CO3	3	2									
	CO4	3	2									
	CO5	3	2									
COURSE CONTENTS:												
MODULE I	INTRODUCTIONS							9 Hours				
Elements of a generalized measurement system- Primary sensing element, variable conversion element, variable manipulation element, data transmission and presentation element; Static and dynamic characteristics; Errors- Types, measurements, remedial methods; Statistical evaluation of measurement data; Calibration – Calibration methodology; Standards-National and International.												

MODULE II	ELCTRICAL AND ELECTRONIC INSTRUMENTS	9 Hours
Measuring instruments- Classification of measuring instruments- Essential requirements of an instrument; Construction and Working Principle of PMMC, MI type instruments, Electro-dynamometer type Wattmeter, Energy Meter; Instrument transformers (CT & PT); Phase sequence indicators; Power factor meters. Digital meters: Electronic multi-meter, Digital voltmeter, Vector Voltmeter.		
MODULE III	DC AND AC BRIDGES	9 Hours
DC bridges –Wheatstone bridge, Kelvin bridge, and their merits and demerits, AC bridges-Maxwell bridge, Anderson bridge, Schering bridge and their merits and demerits; Transformer ratio bridge; Self balancing bridge.		
MODULE IV	STORAGE AND DISPLAY DEVICES	9 Hours
Display devices- LED and LCD display, Comparison between LED and LCDs; Recorders- Strip chart recorders, Single point and Multi-point recorders, X-Y recorders, Magnetic tape recorders; Oscilloscope-CRO, Digital CRO and CRO measurements. Signal Analysis: Wave Analyzer, Spectrum Analyzer& Application of Spectrum Analyzer		
MODULE V	TRANSDUCERS AND DIGITAL DATA ACQUISITION	9 Hours
Transducers- Classification, Characteristics and Selection factors; Passive transducers-Strain cage, RTD and Thermistors transducers; Active transducers-Piezo electric, Hall effect and Thermo electric transducers; A/D and D/A converters; Smart sensors. Digital Data Acquisition System: Interfacing transducers to Electronics Control and Measuring System. Instrumentation Amplifier, Isolation Amplifier.		
TOTAL: 45 HOURS		
LIST OF EXPERIMENTS:		
1. Measure the given low resistance using Kelvin's Double Bridge. 2. Calibration of single Phase Energy Meter 3. Experiment on Temperature/Pressure/Displacement/flow sensors. 4. Perform signal conditioning by using ADC and DAC. 5. Study the displacement transducer using LVDT and obtain its characteristic 6. Simulation Experiment on 7. Measurement of self-inductance by Maxwell's bridge.		
TOTAL: (45+15) 60 HOURS		
REFERENCES:		
1. A.K. Sawhney and Puneet Sawhney, "A Course in Electrical, Electronic Measurements & Instrumentation", Dhanpat Rai and Co., 2012 2. J.B. Gupta, "A Course in Electronic and Electrical Measurements", S.K. Kataria & Sons, Delhi, Jan 2012. 3. H.S. Kalsi, "Electronic Instrumentation", Tata McGraw Hill, 2 nd Edition, 2018. 4. Alan S. Morris, "Principles of Measurements and Instrumentation", 2 nd Edition, Prentice Hall of India, 4. 2003. 5. Murthy D. V. S., "Transducers and Instrumentation", Prentice Hall of India, 13 th Printing, 2018. 6. https://nptel.ac.in/courses/108/105/108105064/		

conservation of biodiversity.		
MODULE IV	NATURAL RESOURCES	10 Hours
Forest resources: Use and over-exploitation, deforestation- timber extraction, mining, dams and their effects on forests and tribal people – Water resources: Use and over utilization of surface and ground water, dams- benefits and problems – Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources – Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity– Energy resources: Growing energy needs, renewable and nonrenewable energy sources, use of alternate energy sources. Energy Conversion processes Biogas – production and uses, anaerobic digestion – Land resources: Land as a resource, land degradation, man induced landslides, soil erosion and desertification – role of an individual in conservation of natural resources – Equitable use of resources for sustainable lifestyles.		
MODULE V	ENVIRONMENTAL POLLUTION	10 Hours
Definition – Source, causes, effects and control measures of: (a) Air pollution (b) Water pollution(c) Soil pollution - soil waste management: causes, effects and control measures of municipal solid wastes – (d) Marine pollution (e) Noise pollution –(f) Nuclear pollution (g) Thermal pollution role of an individual in prevention of pollution.		
TOTAL: 45 HOURS		
LIST OF EXPERIMENTS:		
MINI PROJECT ADDITIONAL TOPICS		
Soil Science		
1. Effects of climate change on soil erosion.		
2. The role of land management in maintaining soil health.		
3. Effects of salinity in coastal region Agricultural activity.		
4. The effects of climate change on agriculture.		
Urban Ecology		
1. How road construction impacts biodiversity and ecosystems.		
2. The effects of urbanization and city planning on water cycles.		
3. Impacts of noise pollution on human health.		
Pollution and Bio-remediation		
1. The role of bio-remediation in removing “forever” chemicals from the environment.		
2. Impacts of air pollution on human health.		
3. How to improve plastic recycling processes?		
4. Individual measures to reduce consumption and creation of micro plastics.		
General Topics		
1. Impact of Urbanization on Local Biodiversity		
2. Renewable Energy Options for Sustainable Living.		
3. Waste Management Strategies in Urban Areas		
4. Climate Change and Its Effects on Local Ecosystems		
5. Air Quality Monitoring in Urban centers		
6. Water Quality Assessment in Local Water Bodies		
7. Green Roof Technology and Its Environmental Benefits		
8. Impact of Plastic Pollution on Marine Life.		
9. Eco-friendly Practices in Agriculture:		
10. The Role of Community Gardens in Urban Sustainability		
11. Alternate energy sources for community Development.		
12. E-Waste Management.		
13. Energy Audit of a building.		
14. Rainwater harvesting system.		
15. Population growth variation among nations.		
16. Population explosion.		
17. Family welfare programme.		

18. Women welfare programme.
19. Child welfare programme.
20. Environmental impact analysis.
21. Role of information technology in environmental protection and human health.
TOTAL: 15 HOURS
REFERENCES:
1. Trivedi.R.K., "Handbook of Environmental Laws, Rules, Guidelines, Compliances and Standards", Vol. I and II, Enviro Media, 3rd edition, BPB publications, 2010.
2. Cunningham, W.P. Cooper, T.H. Gorhani, "Environmental Encyclopedia", Jaico Publ., House, Mumbai, 2001.
3. Dharmendra S. Sengar, "Environmental law", Prentice hall of India PVT LTD, New Delhi, 2007.
4. Rajagopalan, R, "Environmental Studies-From Crisis to Cure", Oxford University Press, 2005.
5. Benny Joseph, "Environmental Science and Engineering", Tata McGraw-Hill, New Delhi, 2006
6. https://en.wikipedia.org/wiki/Carbon_capture_and_storage
7. Ravikrishnan "Environmental Science and Engineering" Sri Krishna Hi-tech Publishing Company Pvt .

7. Load test on single-phase / three-phase transformer.
8. Parallel operation of single-phase transformers.
9. Simulation on Speed control of DC shunt motor (Field control & Armature control method).
10. PLC based DC drives.
TOTAL: 30 HOURS
REFERENCES:
1. Kothari, D.P & Umre, B.S “Laboratory manual for electrical machines”, I.K international Publishing House (P)Ltd. 2 nd Edition, 2017

2302EE352	ANALOG AND DIGITAL ELECTRONICS LABORATORY	L	T	P	C
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PREREQUISITE:

	1. Analog Electronics
	2. Digital Electronics

COURSE OBJECTIVES:

	1. To implement and characterizing the circuit behavior with digital and analog IC"s.
	2. To design and testing of logic gates.
	3. To analyze V-I Characteristics of conventional & special Diodes.
	4. To Design a transistor based amplifier circuits.
	5. To implement and characterizing the circuit behavior with digital and analog IC"s.

COURSE OUTCOMES:

On the successful completion of the course, students will be able to	
CO1	Determine voltage gain from CE and CB configurations.
CO2	Determine the frequency and gain value of various types of oscillators and amplifiers.
CO3	Use simplification techniques to design a combinational hardware circuit.
CO4	Design and implement combinational and sequential circuits.
CO5	Apply various types of biasing and amplifier configuration.

COs Vs. POs MAPPING:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	-	-	-	-	-	-	3	3	-	-
CO2	3	3	3	3	-	-	-	3	3	-	-
CO3	3	3	3	3	-	-	-	3	3	-	-
CO4	3	2	2	1	-	-	-	3	3	-	-
CO5	3	-	-	-	-	-	-	3	3	-	2

COs Vs. PSOs MAPPING:

COs	PSO1	PSO2
CO1	1	3
CO2	1	3
CO3	1	3
CO4	1	3
CO5	1	3

LIST OF EXPERIMENTS:

1. Characteristics of PN junction diode and Zener diode.
2. Characteristics of CE and CB configurations.
3. Design and test a voltage regulator circuit using Zener diode.
4. Design of Hartley Oscillator & Colpitts Oscillator.
5. Design and verify the frequency response of single stage transistor amplifier.
6. Implementation of SISO, SIPO, PISO and PIPO shift registers using Flip-flops.
7. Implementation of Boolean functions, Adder/ Subtractor circuits
8. Design and implementation of Multiplexer and De-multiplexer using logic gates.

9. Design and implementation of encoder and decoder using logic gates.
10. Design and implementation of code converters using logic gates
TOTAL: 30 HOURS
REFERENCES:
1. Mr.K.Nandakumar, and Mr.V.Yokeswaran “Analog and Digital Integrated Circuits Manual”, 2022.
2. D Roy Choudhury and SheilB.Jani, “Linear Integrated Circuits” 4th Edition, New Age International, New Delhi, 2014.
3. RamakantA.Gayakward, “Op-amps and Linear Integrated Circuits”, 4thEdition, PHI Learnings, 2003.
4. Mr.V. Yokeswaran, and Mr.K.Gokulraj —Analog Electronics – Lab Manual 2022.
5. Milman, Halkias and Satyabrata Jit, —Electronic Devices and Circuits 4th Edition, Mc Graw Hill Education (India) Private Ltd, 2015.
6. Integrated circuits: Solution manual: Analog digital circuits and systems manual by Jacob Millman